

What makes a good prediction?

K Arnold

Objectives

- Compare and contrast regression tasks and classification tasks, and give examples of each
- Identify two different ways of measuring accuracy for regression and for classification
- Identify several reasons why a model may predict better on some subsets of data than others

Types of Tasks

- regression
- classification

What makes a good prediction? *Regression*

- **residual**: actual minus predicted
 - If home sold for \\$200k but we predicted \\$250k, residual is __
- (statistical) *bias*: overpredicts vs underpredicts
- absolute error
- squared error
- normalized squared error: MSE / Variance
 - The confusingly named "R-two": $1 - \text{normalized squared error}$

What makes a good prediction? *Classification*

	Email is spam	Email is not spam
Email labeled spam	True positive	False positive (Type 1 error)
Email labeled not spam	False negative (Type 2 error)	True negative

What makes a good prediction? *Classification*

	Email is spam	Email is not spam
Email labeled spam	True positive	False positive (Type 1 error)
Email labeled not spam	False negative (Type 2 error)	True negative

- **Accuracy** (% correct) = $(TP + TN) / (\# \text{ emails})$
- **False negative** ("miss") **rate** = $FN / (\# \text{ true spam})$
- **False positive** ("false alarm") **rate** = $FP / (\# \text{ true not-spam})$

What makes a good prediction? *Classification*

	Email is spam	Email is not spam
Email labeled spam	True positive	False positive (Type 1 error)
Email labeled not spam	False negative (Type 2 error)	True negative

- **Accuracy** (% correct) = $(TP + TN) / (\# \text{ emails})$
- **False negative** ("miss") **rate** = $FN / (\# \text{ true spam})$
- **False positive** ("false alarm") **rate** = $FP / (\# \text{ true not-spam})$
- **Sensitivity** ("true positive rate") = $TP / (\# \text{ true spam})$
 - Sensitivity = $1 - \text{False negative rate}$
- **Specificity** ("true negative rate") = $TN / (\# \text{ true spam})$
 - Specificity = $1 - \text{False positive rate}$
- **Wikipedia article**

If you were designing a spam filter, would you want sensitivity and specificity to be high or low? What are the trade-offs associated with each decision?

Validation

Example: perfectly predicting home price from irrelevant predictors